

Question			Marking details		Marks available					
				AO1	AO2	AO3	Total	Maths	Prac	
7	(a)	(i)	$1.00 \sin \theta_{\text{air}} = 1.60 \sin 30^\circ$ or equivalent or by implication (1) $\theta_{\text{air}} = 53^\circ$ (1) Beam bends sharply to the left at the surface (1)	1	1 1		3	2		
		(ii)	$\theta = \theta_c$ or by implication (1) $\sin \theta = \frac{1}{1.60}$ [=0.625] or equivalent (eg $\theta_c = 39^\circ$) (1) $x = 62.5 / 63$ mm (1)	1 1	1		3	2		
	(b)		Marking points A1. Light [travelling at small angles to axis] hitting core/cladding boundary is totally internally reflected A2. [So] transmitted along fibre without loss B3. Light paths at different angles [to axis] are of different lengths [for given length of fibre] B4. [So] take [slightly] different times and [so] [each] pulse spread out [over time] on arrival at far end C5. Pulses may overlap (if in rapid sequence) C6. Spreading [accept mm dispersion] increases with length of fibre, [so] overlap more likely if fibre longer 5-6 marks <i>Expect at least 4 points made from all of sections A, B and C</i> <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks <i>Expect at least 3 points made from at least two sections from A, B or C</i> <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	6			6			